

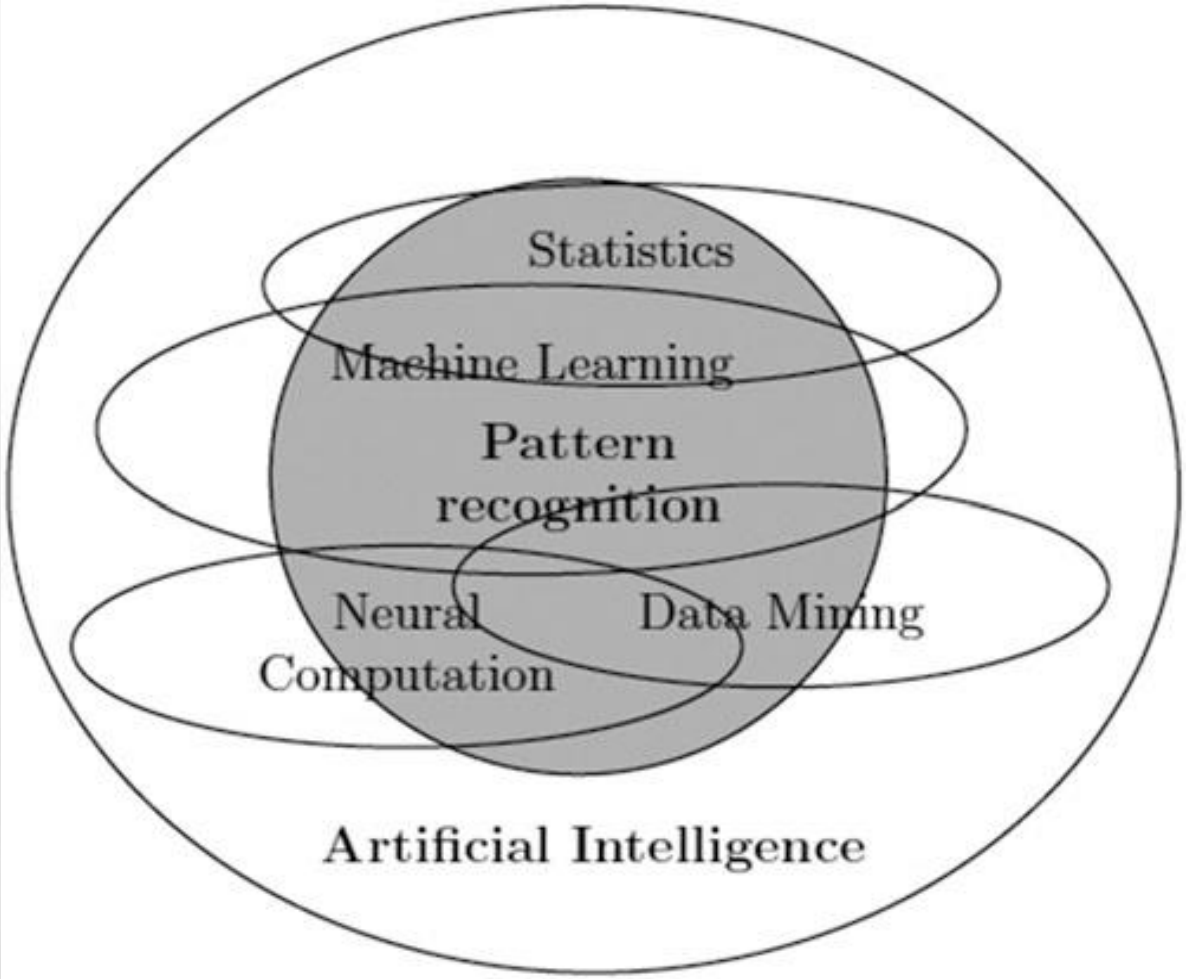
INT 422
Pattern Recognition
2024

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Lecture 11-12: Selected Topics Agenda

- Introduction
- Feature selection
- PCA
- Measurement of system performance

Pattern Recognition and Related Fields



Pattern Recognition System

- **Data acquisition and sensing**:
  - Measurements of physical variables.
  - Important issues: bandwidth, resolution, sensitivity, distortion, latency, etc.
- **Pre-processing**:
  - Removal of noise in data.
  - Isolation of patterns of interest from the background.
- **Feature extraction**:
  - Finding a new representation in terms of features. (ex. Wavelet, contourlet, ..etc)

Pattern Recognition System

- **Model learning and estimation**:
  - Learning a mapping between features and pattern groups and categories.
- **Classification**:
  - Using features and learned models to **assign** a pattern to a category.
- **Post-processing**:
  - Evaluation of confidence in decisions.
  - Exploitation of context to improve performance.
  - Combination of experts.

Feature Selection

- When building a pattern recognition model in real-life, it's almost rare that all the variables in the dataset are useful to build a model.
- Adding redundant variables **reduces** the model's generalization capability and may also reduce the overall **accuracy** of a classifier.
- Furthermore adding more and more variables to a model **increases** the overall **complexity** of the model.
- The goal of feature selection techniques in machine learning is to find the best set of features that allows one to build useful models of studied phenomena.

Principal Components Analysis

Covariance

- **Variance** and **Covariance** are a measure of the "spread" of a set of points around their center of mass (mean).
- **Variance** – measure of the deviation from the mean for points in one dimension e.g. heights
- **Covariance** as a measure of how much each of the dimensions vary from the mean with respect to each other.
- **Covariance is measured between 2 dimensions to see if there is a relationship between the 2 dimensions e.g. number of hours studied & marks obtained.**
- The covariance between one dimension and itself is the variance

Covariance

$$\text{covariance}(X,Y) = \frac{\sum_{i=1}^{n} (X_i - \overline{X})(Y_i - \overline{Y})}{(n-1)}$$

- So, if you had a 3-dimensional data set (x,y,z), then you could measure the covariance between the x and y dimensions, the y and z dimensions, and the x and z dimensions. Measuring the covariance between x and x , or y and y , or z and z would give you the variance of the x , y and z dimensions respectively.

Covariance Matrix

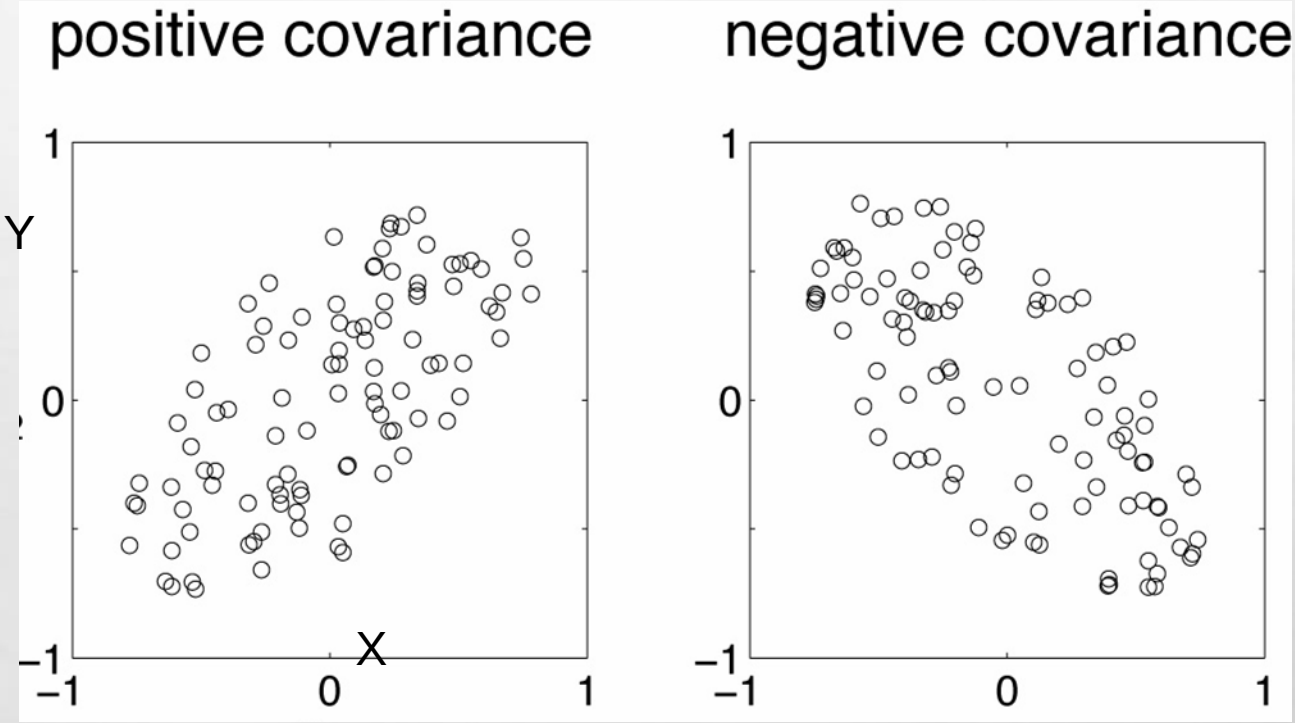
- Representing Covariance between dimensions as a matrix e.g. for 3 dimensions:

$$C = \begin{bmatrix} cov(x,x) & cov(x,y) & cov(x,z) \\ cov(y,x) & cov(y,y) & cov(y,z) \\ cov(z,x) & cov(z,y) & cov(z,z) \end{bmatrix}$$

Variances

- Diagonal is the variances of x, y and z
- cov(x,y) = cov(y,x) hence matrix is symmetrical about the diagonal
- N-dimensional data will result in NxN covariance matrix

Covariance examples



Covariance

- Exact value is not as important as it's sign.
- A positive value of covariance indicates both dimensions increase or decrease together e.g. as the number of hours studied increases, the marks in that subject increase.
- A negative value indicates while one increases the other decreases, or vice-versa e.g. active social life at SU vs performance in CS dept.
- If covariance is zero: the two dimensions are independent of each other e.g. heights of students vs the marks obtained in a subject

Covariance

- Why bother with calculating covariance when we could just plot the 2 values to see their relationship?

  Covariance calculations are used to find relationships between dimensions in high dimensional data sets (usually greater than 3) where visualization is difficult.

principal components analysis (PCA)

- A technique that can be used to simplify a dataset.
- It is a linear transformation that chooses a new coordinate system for the data set such that the **greatest variance** by any projection of the data set comes to lie on the first axis (then called the first principal component),

  the second greatest variance on the second axis, and so on.
- PCA can be used for reducing dimensionality by eliminating the later principal components.

PCA Toy Example

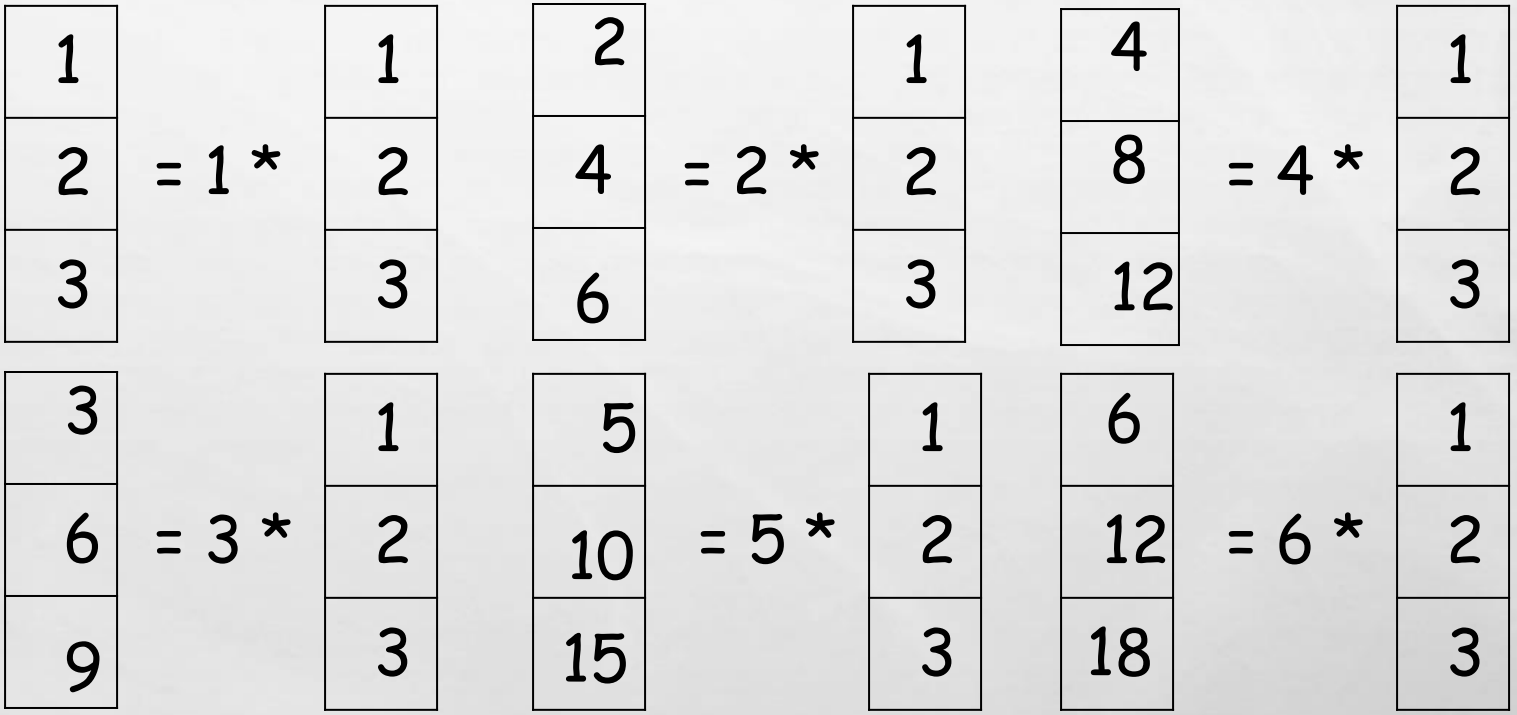
Consider the following 3D points

1	2	4	3	5	6
2	4	8	6	10	12
3	6	12	9	15	18

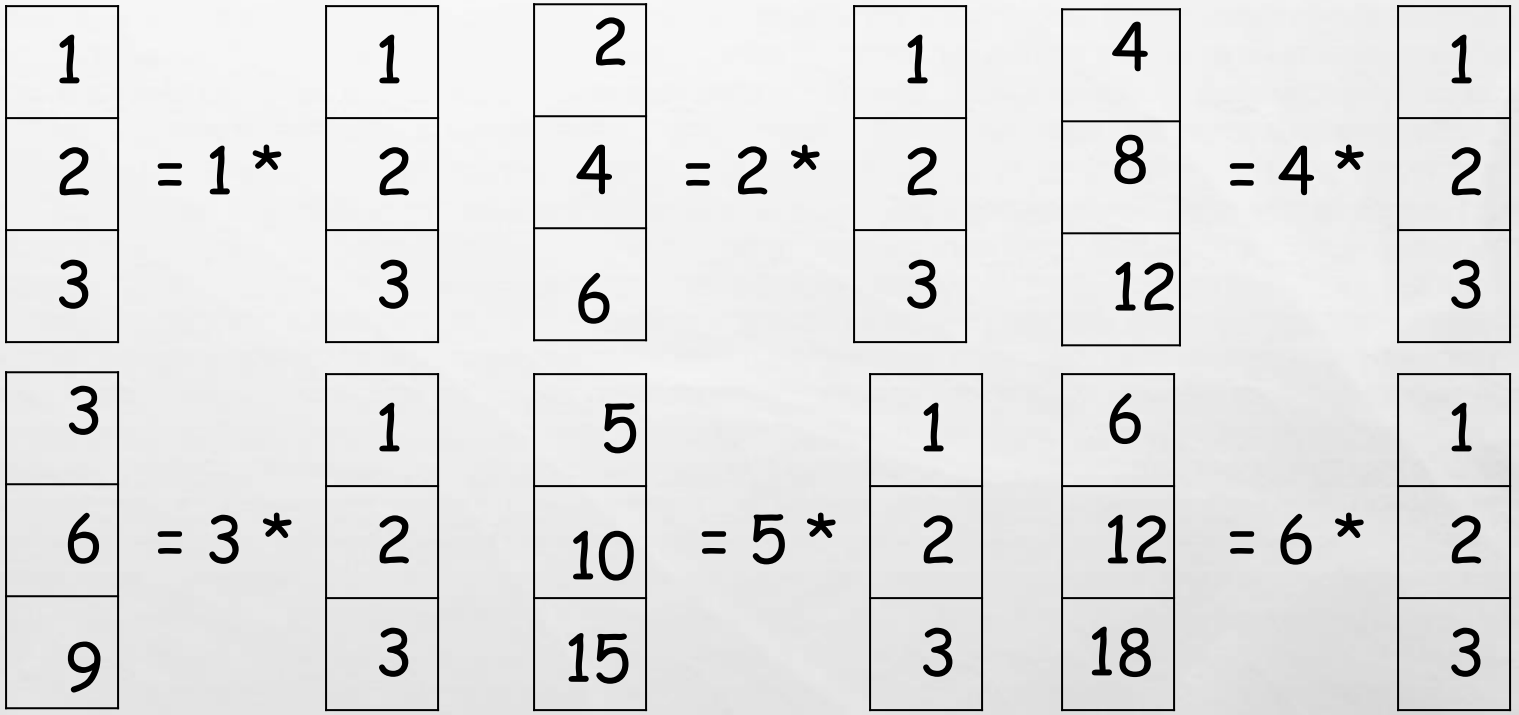
If each component is stored in a byte, we need 18 = 3 x 6 bytes

PCA Toy Example

Looking closer, we can see that all the points are related geometrically: they are all the same point, scaled by a factor:



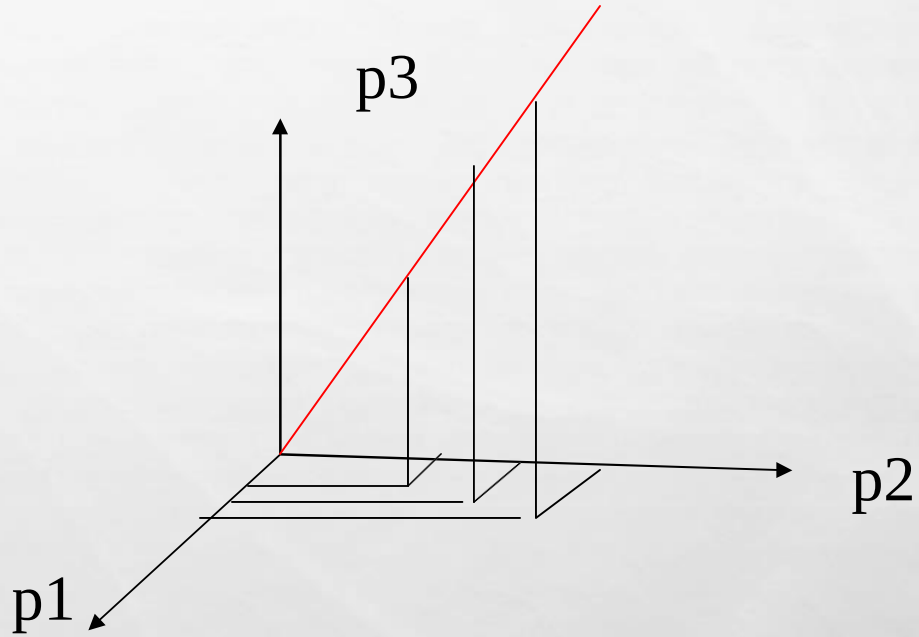
PCA Toy Example



They can be stored using only 9 bytes (50% savings!): Store one point (3 bytes) + the multiplying constants (6 bytes)

Geometrical Interpretation:

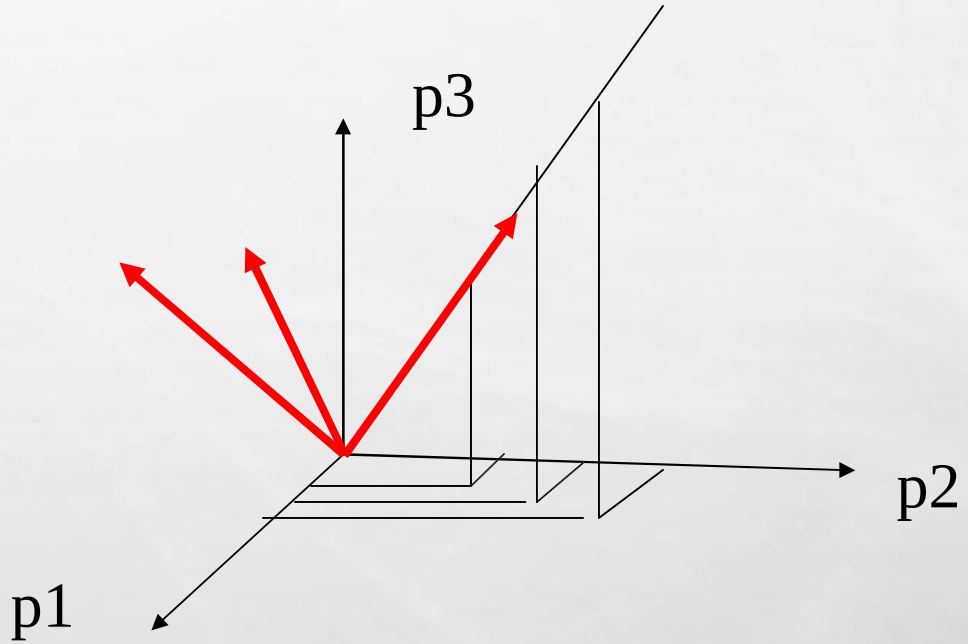
View each point in 3D space.



But in this example, all the points happen to belong to a line:
a 1D subspace of the original 3D space.

Geometrical Interpretation:

Consider a new coordinate system where one of the axes is along the direction of the line:



In this coordinate system, every point has only one non-zero coordinate: we *only* need to store the direction of the line (a 3 bytes image) and the non- zero coordinate for each of the points (6 bytes).

Principal Component Analysis (PCA)

- Given a set of points, how do we know if they can be compressed like in the previous example?
  - The answer is to look into the correlation between the points
  - The tool for doing this is called PCA

PCA

- By finding the eigenvalues and eigenvectors of the covariance matrix, we find that the **eigenvectors with the largest eigenvalues correspond to the dimensions that have the strongest correlation in the dataset**.
- This is the principal component.
- PCA is a useful statistical technique that has found application in:
  - fields such as face recognition and image compression
  - finding patterns in data of high dimension.

PCA Theorem

Let x_1 x_2 … x_n be a set of n N x 1 vectors and let $\bar{x}$ be their average:

$$\mathbf{x}_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{iN} \end{bmatrix} \qquad \bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^{i=n} \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{iN} \end{bmatrix}$$

PCA Theorem

Let X be the N x n matrix with columns $x_1 - x, x_2 - x, \ldots x_n - x$:

$$X = \begin{bmatrix} \mathbf{x}_1 - \bar{\mathbf{x}} & \mathbf{x}_2 - \bar{\mathbf{x}} & \cdots & \mathbf{x}_n - \bar{\mathbf{x}} \end{bmatrix}$$

Note: subtracting the mean is equivalent to translating the coordinate system to the location of the mean.

PCA Theorem

Let Q = X X^T be the N x N matrix:

$$Q = XX^T = \left[\begin{array}{cccc} \mathbf{x}_1 - \bar{\mathbf{x}} & \mathbf{x}_2 - \bar{\mathbf{x}} & \cdots & \mathbf{x}_n - \bar{\mathbf{x}} \end{array} \right] \left[\begin{array}{c} (\mathbf{x}_1 - \bar{\mathbf{x}})^T \\ (\mathbf{x}_2 - \bar{\mathbf{x}})^T \\ \vdots \\ (\mathbf{x}_n - \bar{\mathbf{x}})^T \end{array} \right]$$

Notes:

1. Q is square
2. Q is symmetric
3. Q is the *covariance* matrix [aka scatter matrix]
4. Q can be very large (in vision, N is often the number of pixels in an image!)

PCA Theorem

Theorem:

Each x_j can be written as:

$$\mathbf{x}_j = \bar{\mathbf{x}} + \sum_{i=1}^{i=n} g_{ji}\mathbf{e}_i$$

where e_i are the n eigenvectors of Q with non-zero eigenvalues.

Notes:

1. The eigenvectors e_1 e_2 … e_n span an ***eigenspace***
2. e_1 e_2 … e_n are N x 1 orthonormal vectors (directions in N-Dimensional space)
3. The scalars g_{ji} are the coordinates of x_j in the space.

$$g_{ji} = (\mathbf{x}_j - \bar{\mathbf{x}}).\mathbf{e}_i$$

Using PCA to Compress Data

- Expressing x in terms of $e_1 \ldots e_n$ has not changed the size of the data
- However, if the points are highly correlated many of the coordinates of x will be zero or closed to zero.

note: this means they lie in a lower-dimensional linear subspace

Using PCA to Compress Data

- Sort the eigenvectors e_i according to their eigenvalue:

$$\lambda_1 \geq \lambda_2 \geq \ldots \lambda_n$$

- Assuming that $\lambda_i \approx 0 \text{ if } i > k$

- Then

$$\mathbf{x}_j \approx \bar{\mathbf{x}} + \sum_{i=1}^{i=k} g_{ji}\mathbf{e}_i$$

PCA Example –STEP 1

http://kybele.psych.cornell.edu/~edelman/Psych-465-Spring-2003/PCA-tutorial.pdf

- DATA:

x	y
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
3.1	3.0
2.3	2.7
2	1.6
1	1.1
1.5	1.6
1.1	0.9

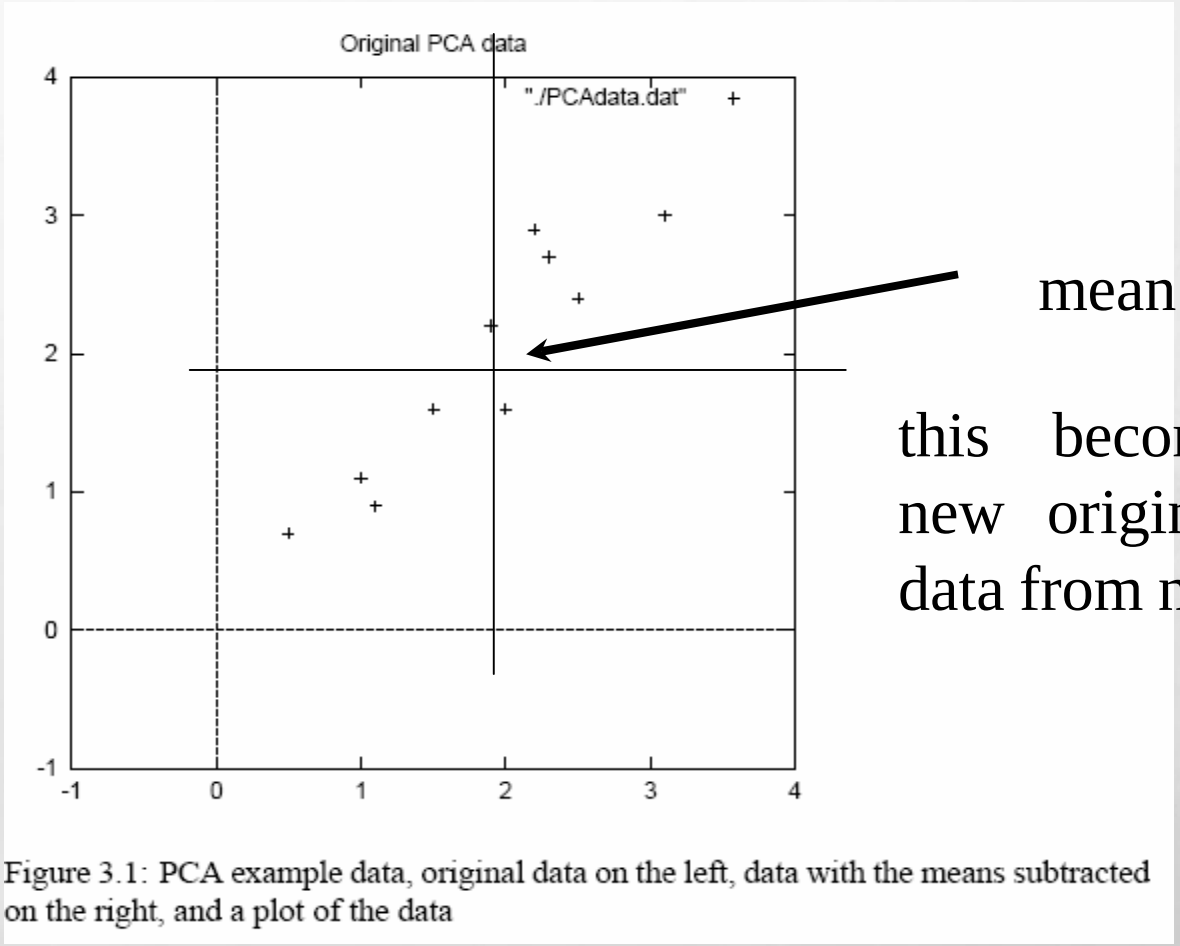


Figure 3.1: PCA example data, original data on the left, data with the means subtracted on the right, and a plot of the data

PCA Example –STEP 2

- Calculate the covariance matrix

$$cov = \begin{pmatrix} .616555556 & .615444444 \\ .615444444 & .716555556 \end{pmatrix}$$

- since the non-diagonal elements in this covariance matrix are positive, we should expect that both the x and y variable increase together.

PCA Example –STEP 3

- Calculate the eigenvectors and eigenvalues of the covariance matrix

$$\text{Eigenvalues} = \begin{pmatrix} .0490833989 \\ 1.28402771 \end{pmatrix}$$

$$\text{eigenvectors} = \begin{pmatrix} -.735178656 & -.677873399 \\ .677873399 & -735178656 \end{pmatrix}$$

PCA Example –STEP 3

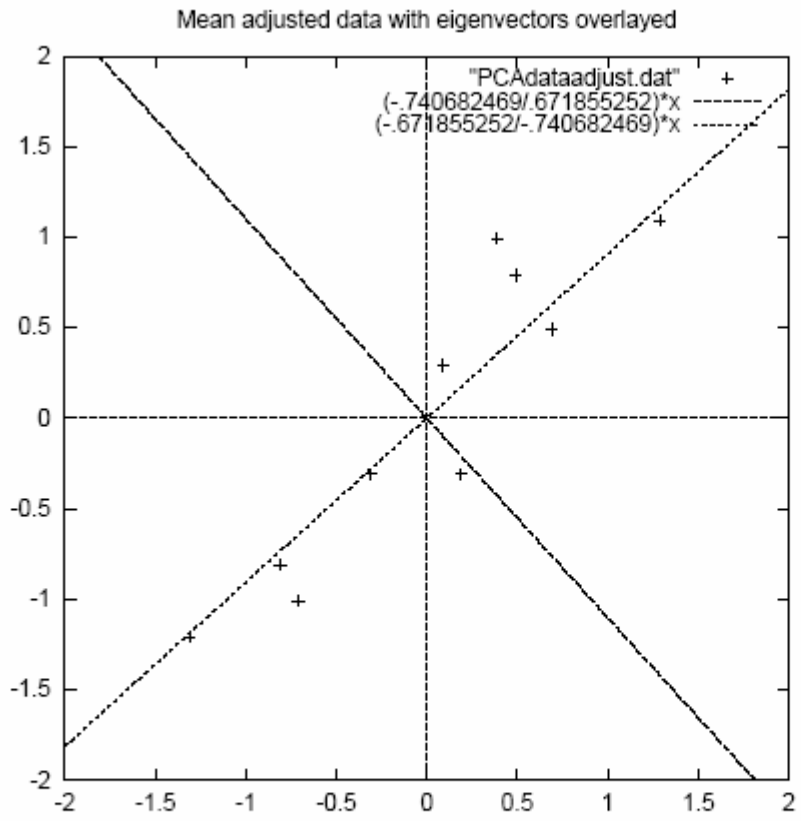


Figure 3.2: A plot of the normalised data (mean subtracted) with the eigenvectors of the covariance matrix overlayed on top.

- eigenvectors are plotted as diagonal dotted lines on the plot.
- Note they are perpendicular to each other.
- Note one of the eigenvectors goes through the middle of the points, like drawing a line of best fit.
- The second eigenvector gives us the other, less important, pattern in the data, that all the points follow the main line, but are off to the side of the main line by some amount.

PCA Example –STEP 4

- Feature Vector

  FeatureVector = (eig_1 eig_2 eig_3 … eig_n)

  We can either form a feature vector with both of the eigenvectors:

$$\begin{pmatrix} -.677873399 & -.735178656 \\ -.735178656 & .677873399 \end{pmatrix}$$

  or, we can choose to leave out the smaller, less significant component and only have a single column:

$$\begin{pmatrix} -.677873399 \\ -.735178656 \end{pmatrix}$$

PCA Example –STEP 5

- Deriving new data coordinates

FinalData = RowFeatureVector x RowZeroMeanData

RowFeatureVector is the matrix with the eigenvectors in the columns *transposed* so that the eigenvectors are now in the rows, with the most significant eigenvector at the top

RowZeroMeanData is the mean-adjusted data *transposed*, ie. the data items are in each column, with each row holding a separate dimension.

Note: his is essential Rotating the coordinate axes so higher-variance axes come first.

PCA Example –STEP 5

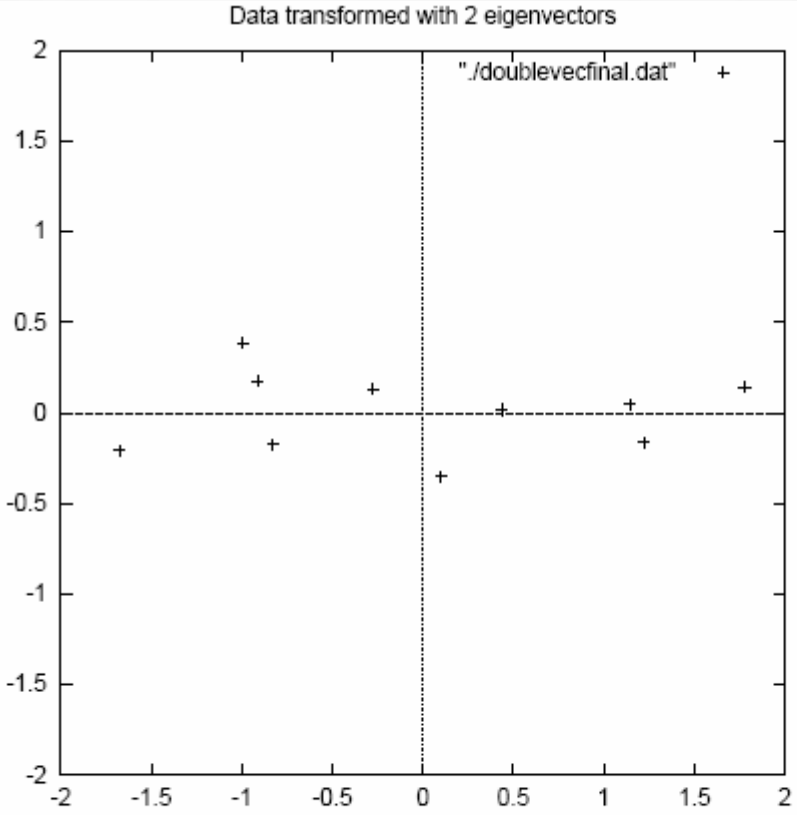
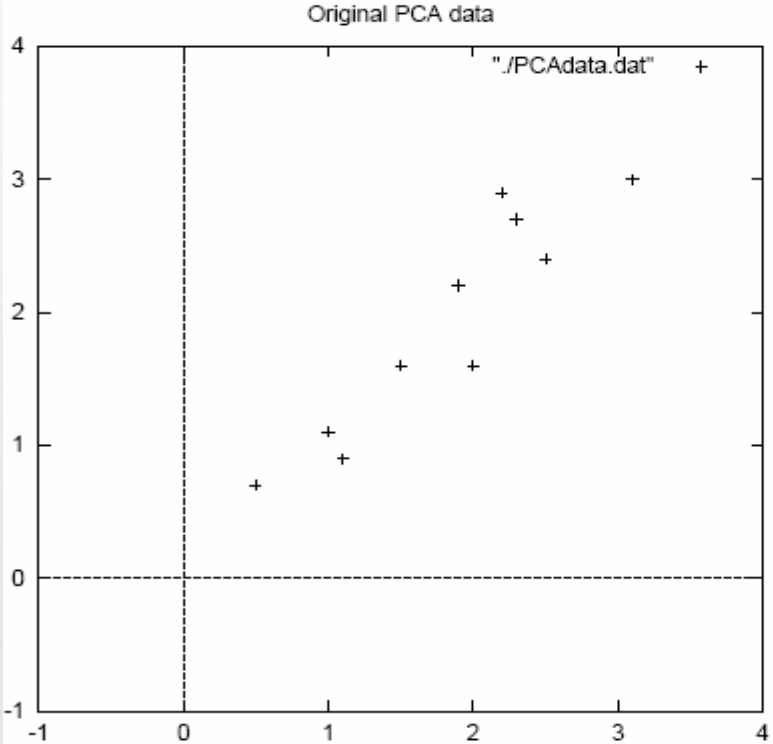


Figure 3.3: The table of data by applying the PCA analysis using both eigenvectors, and a plot of the new data points.

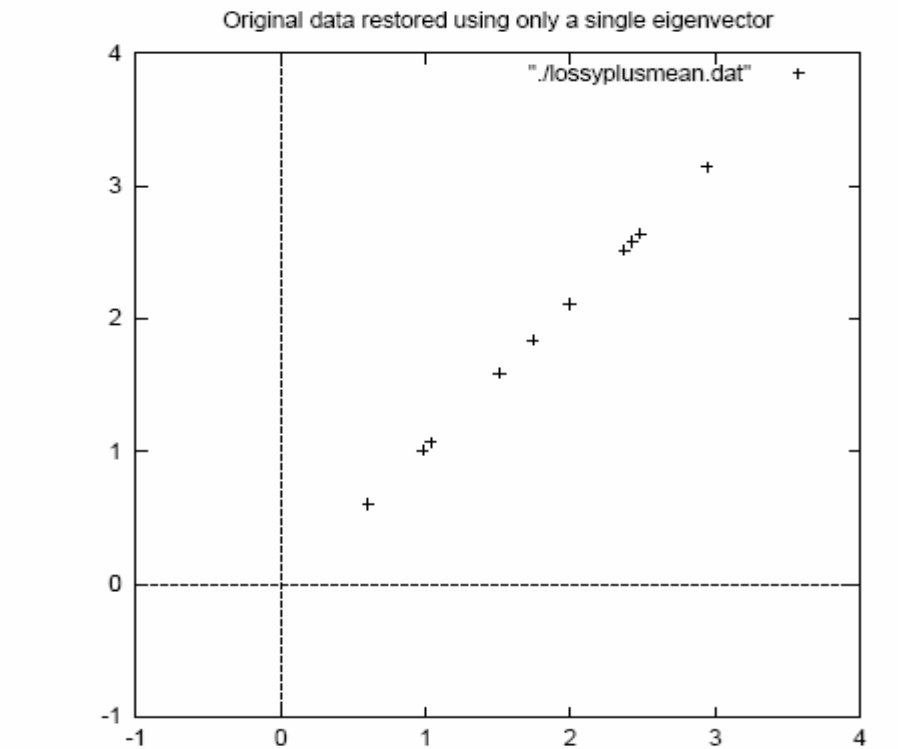
PCA Example : Approximation

- If we reduced the dimensionality, obviously, when reconstructing the data we would lose those dimensions we chose to discard. In our example let us assume that we considered only the x dimension.

PCA Example : Final Approximation



2D point cloud



Approximation using one eigenvector basis

Introduction
Performance Measurement

- An important element of any machine learning workflow is the evaluation of the performance of the model.
- This is the process where we use the trained model to make predictions on previously unseen, labelled data.
- In the case of classification, we then evaluate how many of these predictions the model got right.

Metrics

It is extremely important to use **quantitative metrics** for evaluating a machine learning model

- Until now, we relied on the **cost function value** for regression and classification
- Other metrics can be used to **better evaluate** and understand the model
- **For classification**
  - ✓ Accuracy/Precision/Recall/F1-score, ROC curves,…
- **For regression**
  - ✓ Normalized RMSE, Normalized Mean Absolute Error (NMAE),…

Performance Measurement
1. Accuracy

- The overall **accuracy** of a model is simply the number of correct predictions divided by the total number of predictions. An accuracy score will give a value between 0 and 1, a value of 1 would indicate a perfect model.

$$accuracy = CorrectPredictions/TotalPredictions$$

- This metric should rarely be used in isolation, as on imbalanced data, where one class is much larger than another, the accuracy can be highly misleading.

Performance Measurement 1. Accuracy (cont.,)

- If a cancer example is used. Imagine we have a dataset where only 1% of the samples are cancerous. A classifier that simply predicts all outcomes as benign would achieve an accuracy score of 99%.
- However, this model would, in fact, be useless and dangerous as it would never detect a cancerous observation.

Performance Measurement
2. Confusion Matrix

A **confusion matrix** is an extremely useful tool to observe in which way the model is wrong (or right!). It is a matrix that compares the number of predictions for each class that are correct and those that are incorrect.

True positives: The number of positive observations the model correctly predicted as positive.

False-positive: The number of negative observations the model incorrectly predicted as positive.

True negative: The number of negative observations the model correctly predicted as negative.

False-negative: The number of positive observations the model incorrectly predicted as negative.

Performance Measurement
2. Confusion Matrix

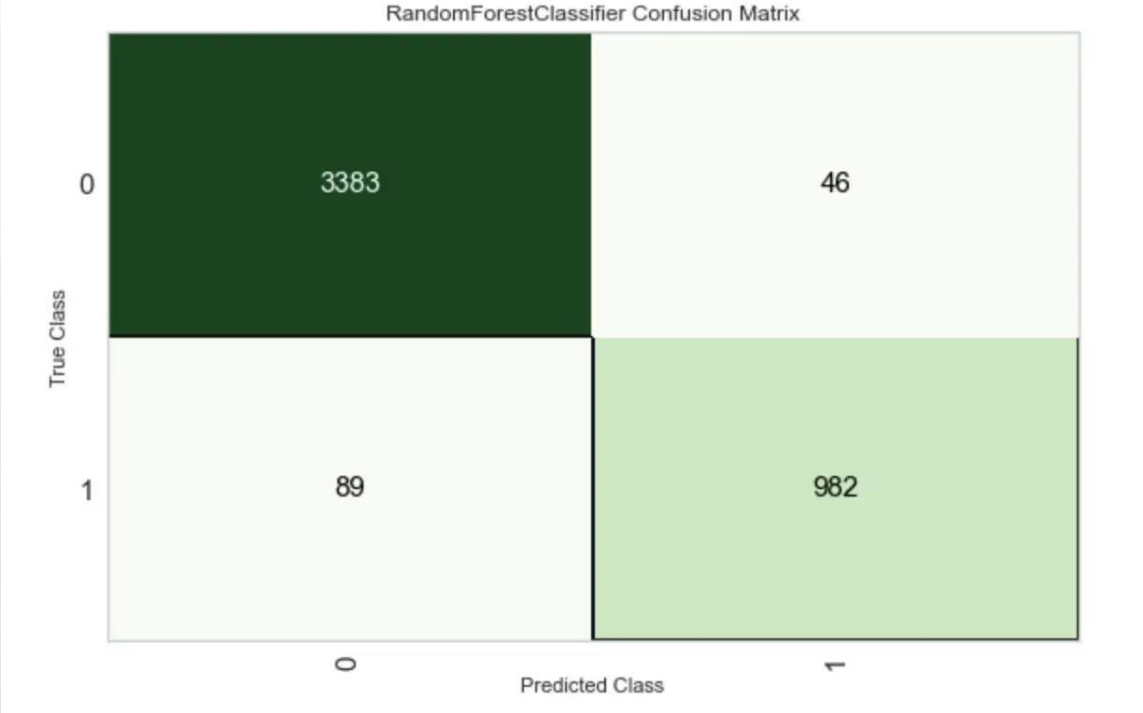
The image below shows a confusion matrix for a classifier. Using this we can understand the following:

•The model correctly predicted 3,383 negative samples but incorrectly predicted 46 as positive.

•The model correctly predicted 962 positive observations but incorrectly predicted 89 as negative.

•We can see from this confusion matrix that the data sample is imbalanced, with the negative class having a higher volume of observations.

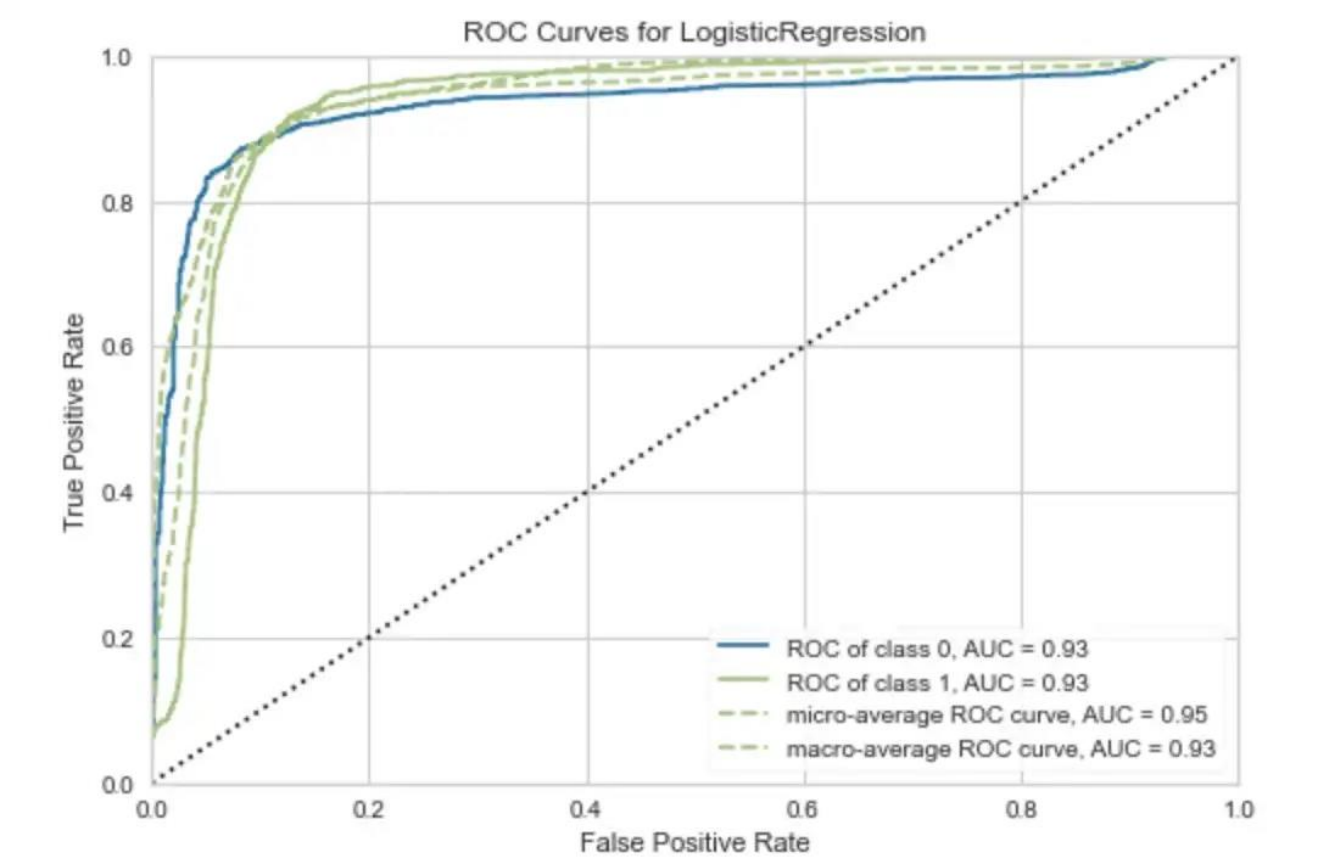
Performance Measurement
2. Confusion Matrix



Performance Measurement
3. AUC/ROC

- A classifier such as logistic regression will return the probability of an observation belonging to a particular class as the prediction output.
- For the model to be useful this is usually converted to a binary value e.g. either the sample belongs to the class or it doesn't.
- To do this a classification threshold is used, for example, we might say that if the probability is above 0.5 then the sample belongs to class 1.
- The **ROC** (Receiver Operating Characteristics) curve is a plot of the performance of the model (a plot of the true positive rate and the false positive rate) at all classification thresholds.
- The **AUC** is the measurement of the entire two-dimensional area under the curve and as such is a measure of the performance of the model at all possible classification thresholds.
- ROC curves plot the accuracy of the model and therefore are best suited to **diagnose the performance of models where the data is not imbalanced.**

Performance Measurement
3. AUC/ROC



Performance Measurement
4. Precision

- Measures how good the model is at correctly identifying the positive class.
- In other words out of all predictions for the positive class how many were actually correct? **Using alone this metric for optimizing a model we would be minimizing the false positives.**
- This might be **desirable for our fraud detection example**, but would be less useful for diagnosing cancer as we would have little understanding of positive observations that are missed.
- **Precision= (TP/(TP+FP))**

Performance Measurement

5. Recall

- tell us how good the model is at correctly predicting all the positive observations in the dataset. However, it does not include information about the false positives so would be more useful in the cancer example.
- **Recall = (TP/(TP+FN))**

Precision and recall

Suppose that $y = 1$ in presence of a **rare class** that we want to detect

Precision *(How much we are precise in the detection)*

Of all patients where we classified $y = 1$, what fraction actually has the disease?

$$\frac{\text{True Positive}}{\text{\# Estimated Positive}} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

Recall *(How much we are good at detecting)*

Of all patients that actually have the disease, what fraction did we correctly detect as having the disease?

$$\frac{\text{True Positive}}{\text{\# Actual Positive}} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

Confusion matrix

Estiamted class \ Actual class	1 (p)	0 (n)
1 (Y)	True positive (TP)	False positive (FP)
0 (N)	False negative (FN)	True negative (TN)

Summary

Balancing sensitivity and specificity is incredibly challenging when it comes to medical applications

The results should **always be validated** with another pool of people

Furthermore, we need to be **concerned of what the model is actually learning:**

- Does the results align with the medical knowledge?
- Was the dataset well representative of the population or there was selection bias?
- Do we accounted for all external factors (confounding) that could interfere with the response?

Review Questions

- **Explain the following:**

1. Why does the process of feature selection is important in pattern recognition?
2. What are the differences between Variance and Covariance? and why they are used?
3. What is principal components analysis (PCA) and why is it used?
4. The different types of Performance Measurement for any pattern recognition system.

References

- https://www.cse.psu.edu/~rtc12/CSE586Spring2010/lectures/pcaLectureShort.pdf

Questions ?

Thank you